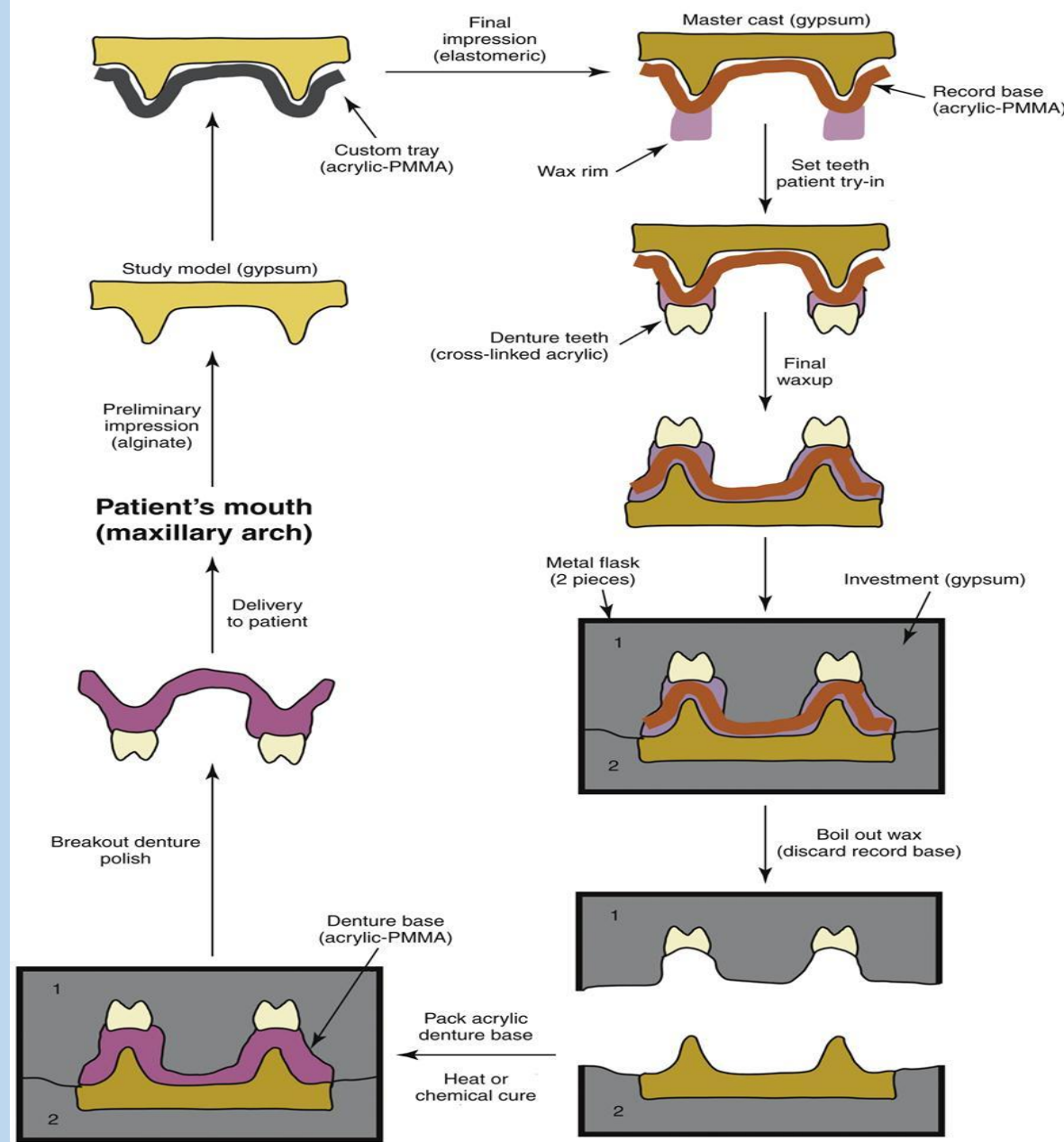


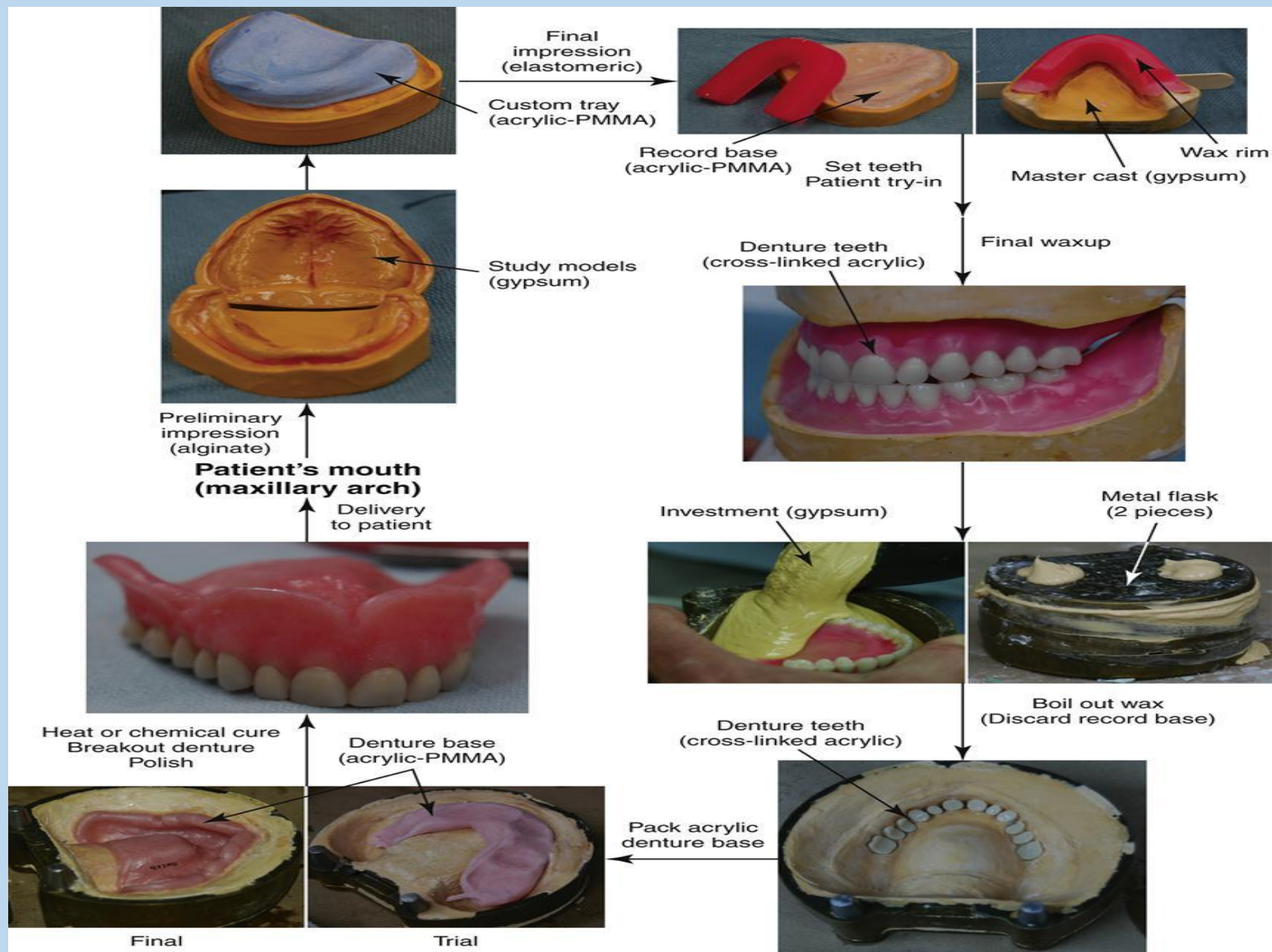


Denture Base Materials (Practical)

Dr Sherif Refat







Manipulation of heat-cured PMMA

I-General instructions

The majority of PMMA formulations are supplied as polymeric powders and colorless monomer liquids.

The exothermic chemical reaction starts upon mixing of the PMMA powder and liquid

To avoid any errors, it is necessary to use the recommended ratio of PMMA powder and liquid (usually 3–3.5:1 v/v).

In the case of a high powder-to-liquid ratio, this means that not all of the PMMA beads will be wet, leading to a granular texture

while a low powder-to-liquid ratio enhances polymerization shrinkage and dimensional changes



II-Stages of manipulation

Depending on the consistency: the manipulation of PMMA mixing can be divided into distinct stages (such as **sandy, stringy, doughy, rubbery, and stiff stages**), which are followed by packing and finishing of denture bases



Sandy

The mixing process starts with the **sandy stage**, giving the mixture a grainy appearance. There is hardly any chemical interaction or polymerization during this stage

Stringy

In the **stringy stage**, the monomer molecules attack and dissolve PMMA particles.

characterized by formation of sticky strings upon touching or stretching

Dough stage is characterized by the loss of stringiness and stickiness

Dough

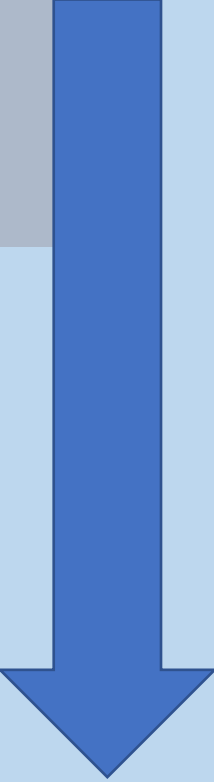
the mixture is considered suitable for packing into a dental flask.

rubbery

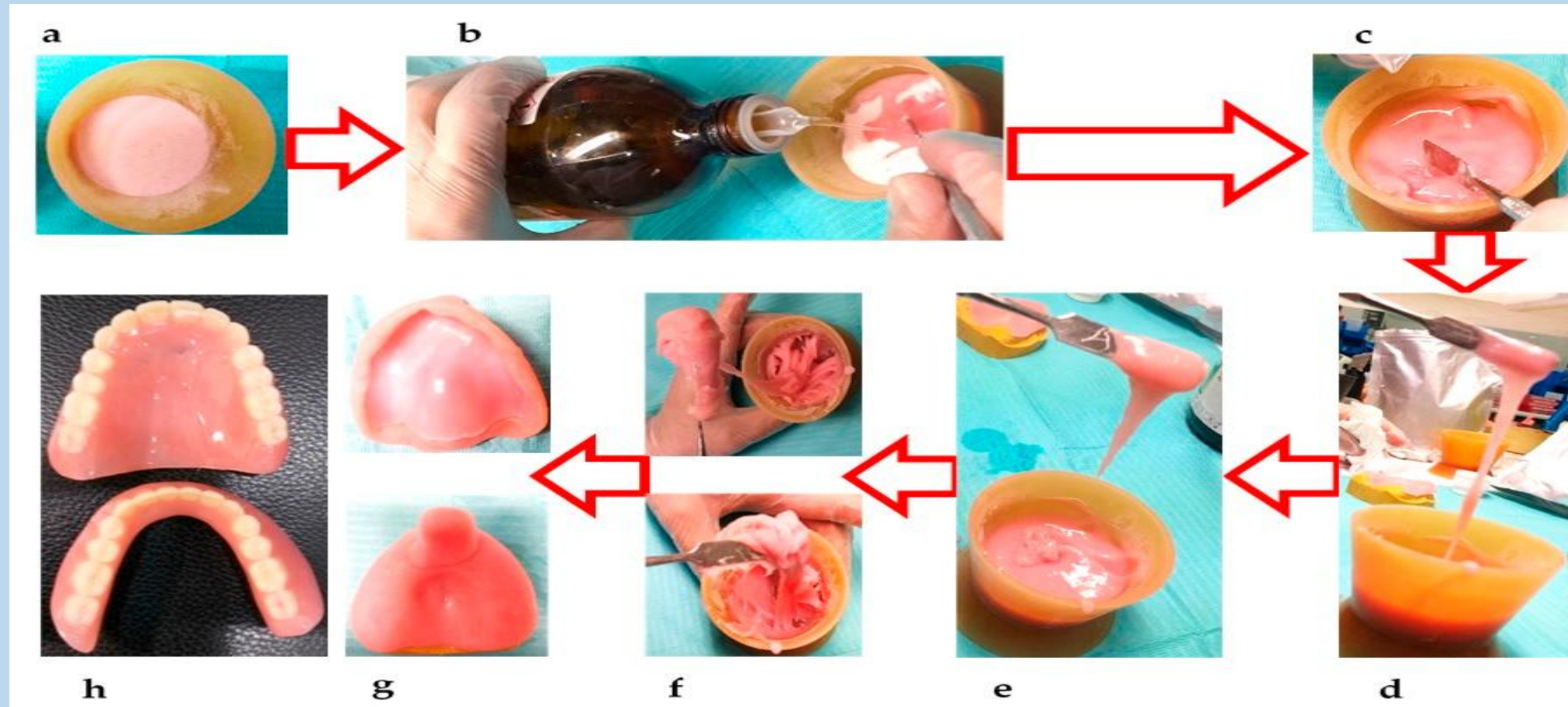
The doughy stage progress to the rubbery stage by further conversion of monomers into polymers and evaporation of the residual monomers.

rubbery appearance of the matrix (which rebounds after releasing compressive or tensile stresses)

Unable to be packed into the compression molds



Stiff Dry, stiff, and resistant to plastic deformation



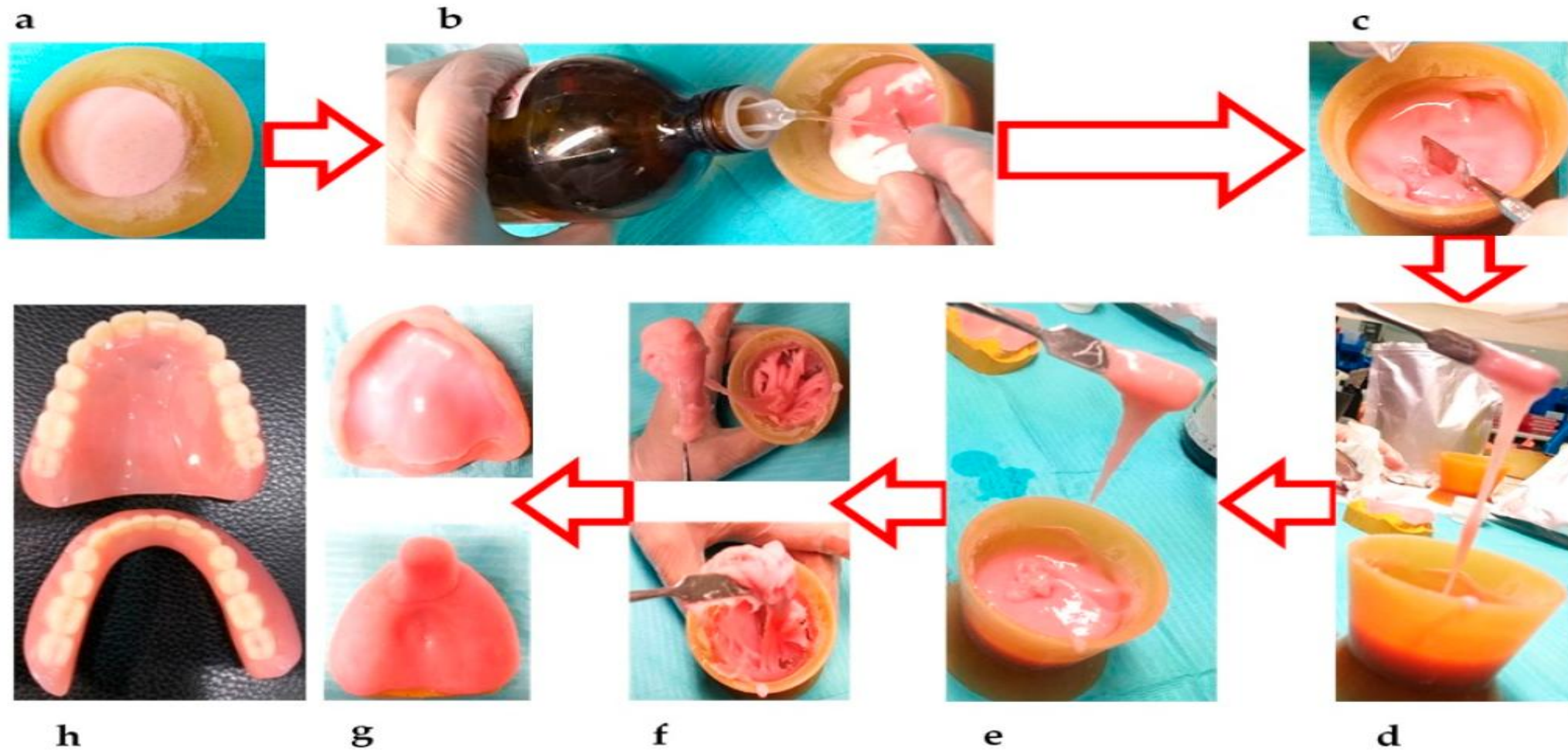


Figure 2. Manipulation and various stages of PMMA manual mixing and handling for denture fabrication: (a) PMMA powder in a mixing bowl; (b) pouring of the monomer solution into PMMA for mixing; (c) sandy stage, whereby the monomer solution has full wetted and saturated the PMMA particles; (d) start of stringy stage; (e) progression of the stringy stage; (f) dough stage ready for packing and plastic molding; (g) rubbery stage followed by the hardened PMMA, whereby plastic deformation is no longer possible; (h) fabricated denture using heat-cured PMMA and acrylic teeth after finishing and polishing.



III-Mixing techniques

- 1- Manual
 - or
 - 2- Ultrasonic
-
- Although the ultrasonic mixing of modified PMMA has been shown to be superior in texture and packing , its use is limited in dental laboratories due to the need of additional equipment and the high cost

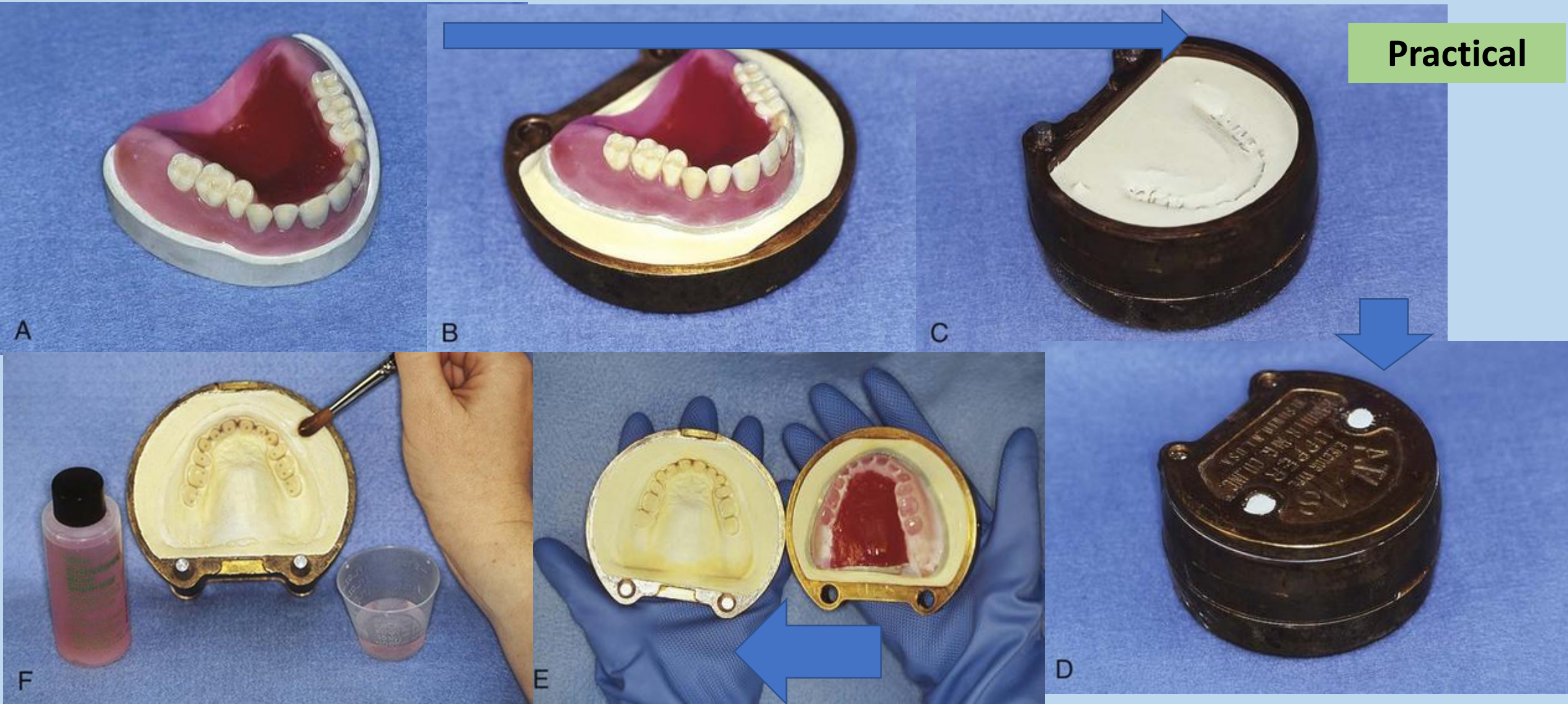
IV-Compression molding packing method

- - the PMMA in the dough stage is packed into specialized dental flasks under **compression**.
- -Due to the method's **accuracy and low cost**, compression molding is frequently used for making **heat-cured PMMA prostheses**.
- -The dental flask packed with PMMA is heated in a **water bath under the defined time and temperature conditions (look at table of heat cycles)**.
- -the temperature of the water bath should be **increased gradually**.
- Immersion of the dental flask directly into boiling water results in evaporation of the monomer (boiling point 100.3 C) before polymerization and increasing porosity in the cured material

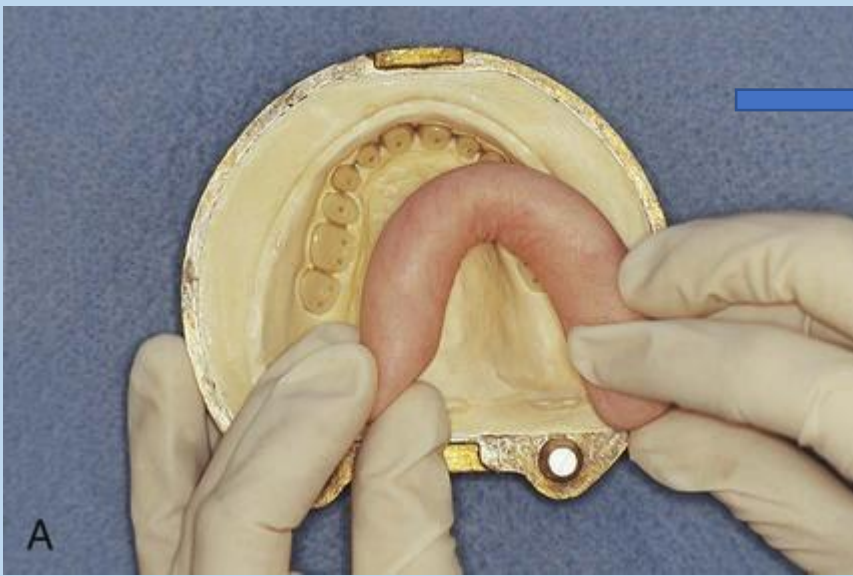


A cross-sectional representation of the denture flask and its contents

Practical



Steps in mold preparation (compression molding technique). A, Completed tooth arrangement prepared for flasking process. B, Master cast embedded in properly contoured dental stone. C, Occlusal and incisal surfaces of the prosthetic teeth are exposed to facilitate subsequent denture recovery. D, Fully flaked complete maxillary denture. E, Separation of flask segments during wax elimination process. F, Placement of alginate-based separating medium.



Steps in resin packing (compression molding technique). **A**, Properly mixed resin is bent into a horseshoe shape and placed into the mold cavity. **B**, The flask assembly is placed into a flask press, and pressure is applied. **C**, Excess material is carefully removed from the flask. **D**, The flask is transferred to a flask carrier, which maintains pressure on the assembly during processing in water bath

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Important notes:

- 1- in mold preparation technique :
- to remove wax and record base the denture flask is immersed in boiling water .
- The flask is then removed from the water and opened. The record base and softened wax remain in the lower portion of the denture flask while the prosthetic teeth remain firmly embedded in the investing stone of the remaining segment.
- The record base and softened wax are carefully removed from the surface of the mold.

2- APPLICATION OF A SEPARATING MEDIUM : separating medium must prevent direct contact between the denture base resin and the mold surface.



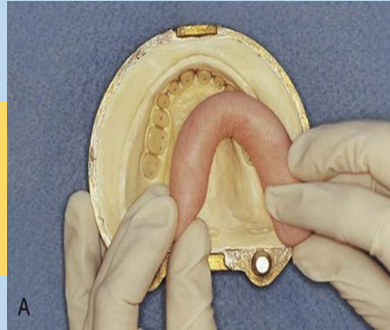
- **Separator**
- should not be permitted to contact exposed portions of acrylic resin teeth, since its presence interferes with chemical bonding between acrylic resin teeth and denture base resins.



3- Usually the P/L “polymer-to-monomer ratio” is of approximately 3 : 1 by volume.

Using a 3 : 1 ratio, the volumetric shrinkage can be limited to approximately 7%.

4- **DOUGH-FORMING TIME** : The time required for the resin mixture to reach a dough-like stage is termed *the dough-forming time*.

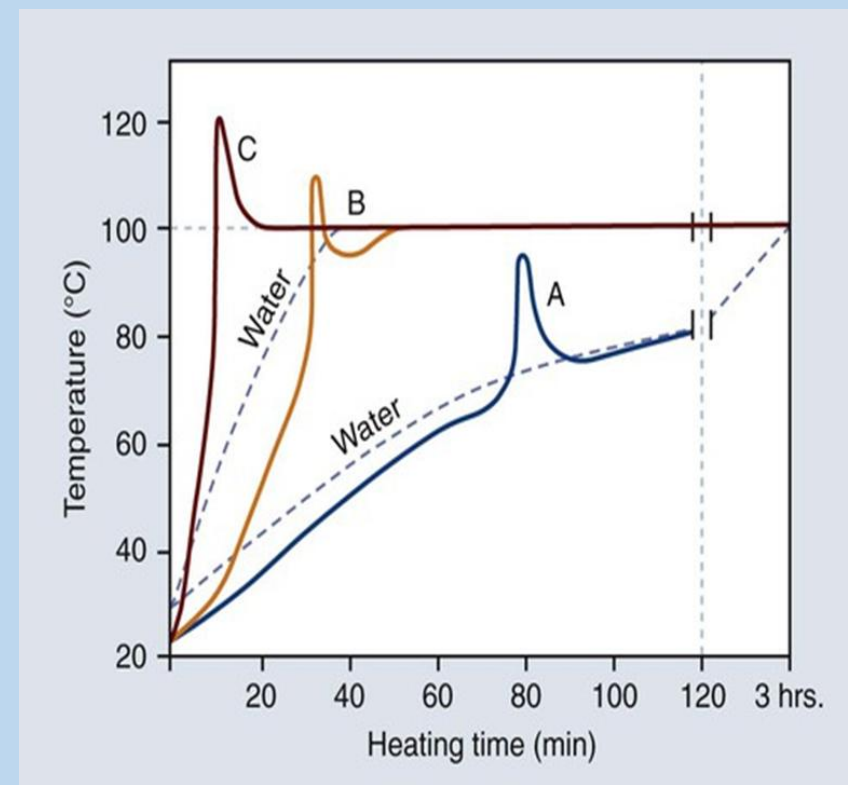
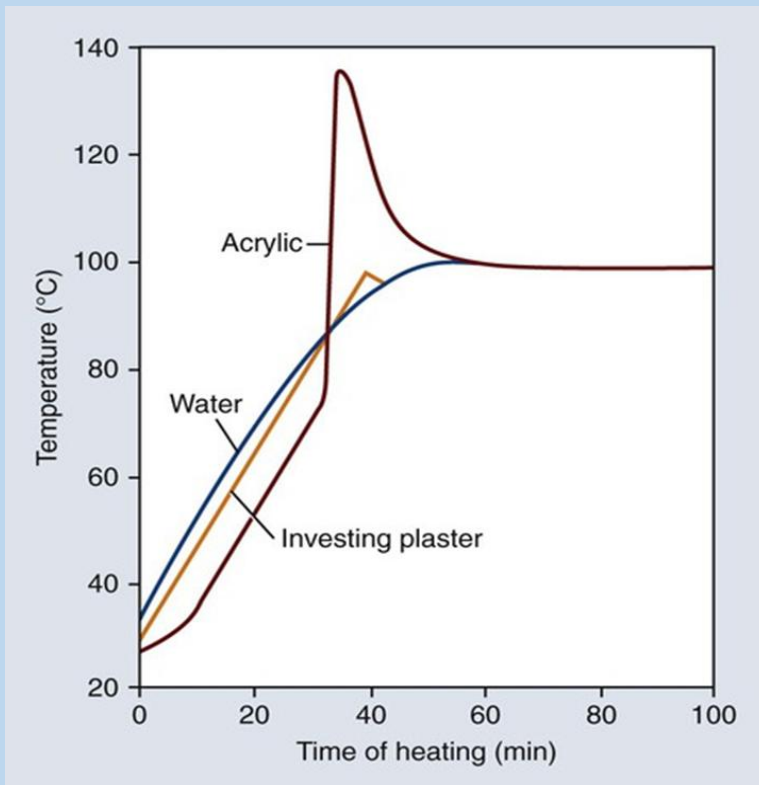


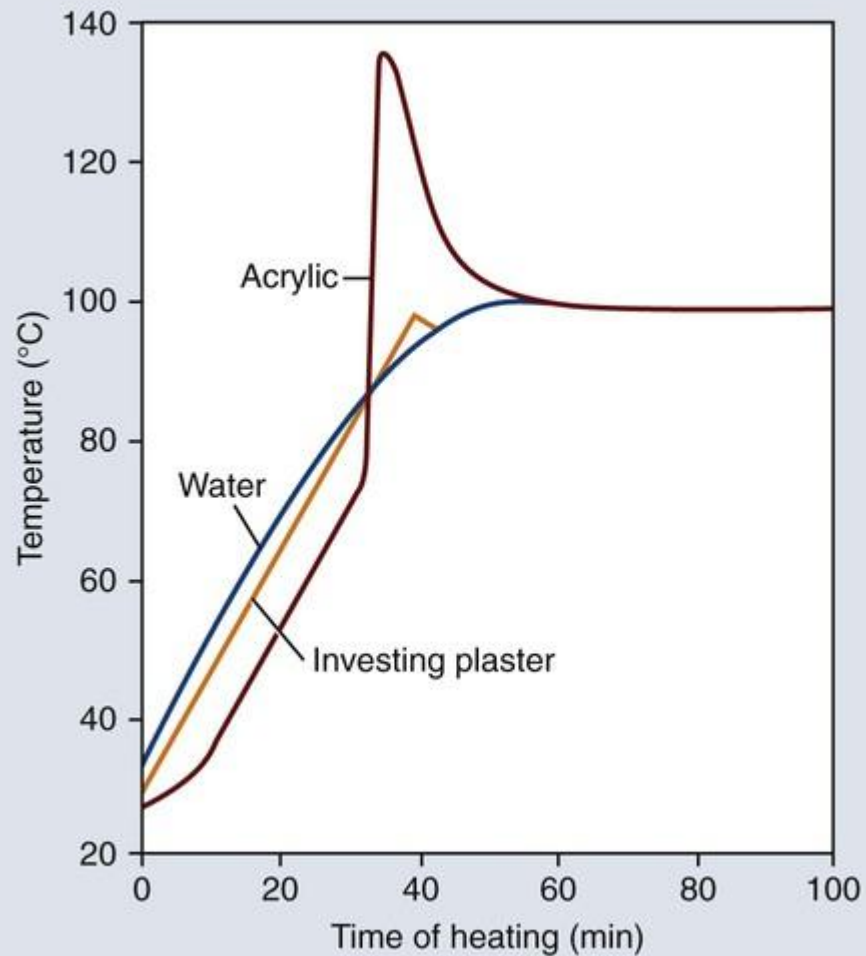
5- **Working time** is defined as the time a denture base material remains in the dough-like stage

7-TEMPERATURE RISE :The polymerization of denture base resins is exothermic

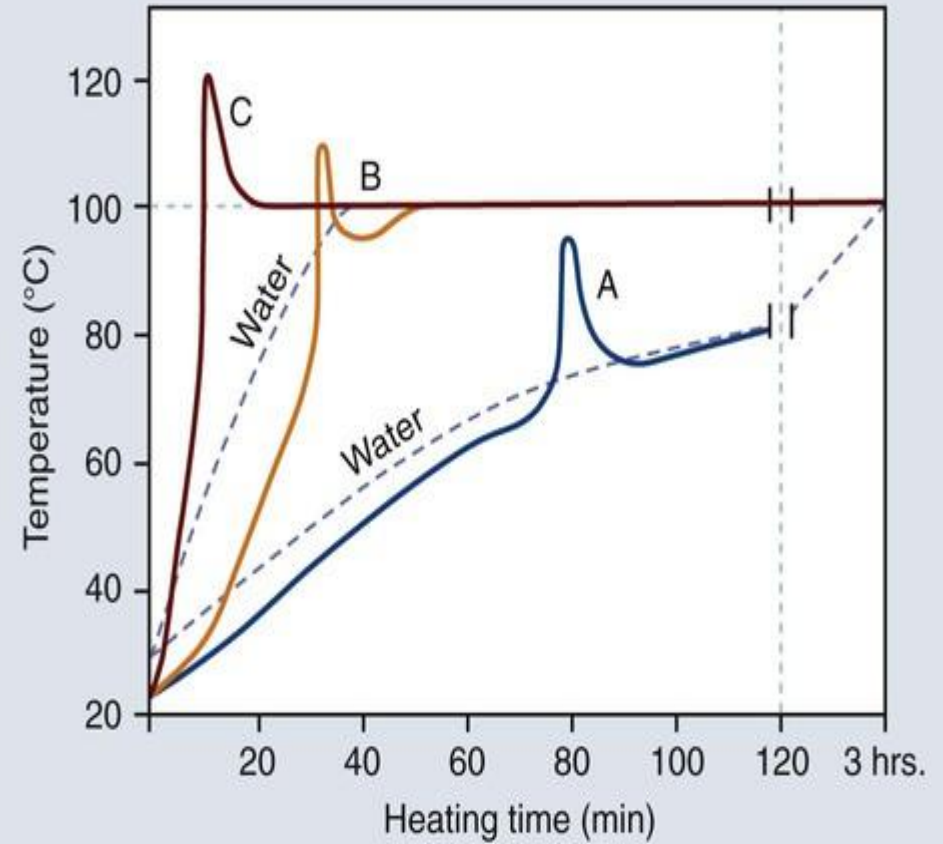
-the resin occupies a position in the center of the mold and therefore heat penetration takes a longer time than water.

- Uncontrolled temperature rise cause unreacted monomer boiling and can cause porosity





Temperature-time heating curves for the water bath, investing plaster, and acrylic resin during the polymerization of a 25.4-mm cube of denture resin.



Temperature changes in acrylic resin when subjected to various curing schedules.

The curing cycle presented here is unsatisfactory because of the marked temperature increase of resin during the early stages of polymerization. Rapid heating of the resin mass may cause the monomer to boil, producing porosity in the polymerized denture base.

Research has led to the development of guidelines for polymerization cycles of denture base resins

Here is an example

Heat Cycles	Temperature	Time	Terminal Boiling
1	74 °C	8 h	None
2	74 °C	8 h	1 h
3	74°C	3 h	1 h
4	73.9 °C	12 h	None
5	70 °C	3 h	1 h
6	100 °C	20 min	None



8- cooling and finishing:- due to the mismatch of the thermal expansion coefficients of PMMA and investment plaster, it is advised to cool down the dental flask to room temperature gradually to avoid distortion of the denture base.



- remove the flask from the water bath to allow bench cooling for half an hour, followed by immersion in tap water (15 min) before opening .



- The PMMA dentures should be finished and polished before use in the oral cavity



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• *THANK YOU*

